

lecture9 Practice Questions and Solutions

Question 1 (q1_sol.py)

Write a function called `print_in_box(word)` that prints `word` in a box as shown. Use the `print_inBars` function from the previous question to print the middle line of the box. Then ask the user to enter a movie name and print in a box.

```
Enter a movie name: The Matrix
+-----+
|The Matrix|
+-----+
```

Sample Solution

```
def print_inBars(word):
    print(f'|{word}|')

def print_in_box(word):
    bar = '+' + '-' * len(word) + '+'
    print(bar)
    print_inBars(word)
    print(bar)

movie = input("Enter a movie name: ")
print_in_box(movie)
```

Question 2 (q2_sol.py)

Write a program that asks the user to enter an int that is 0 or greater, and a word, and then prints the word in a numbered list as shown. Use a for-loop in your solution. No functions are required, but you can make one if you want.

```
Enter an int: 3
Enter a word: hello!
hello!
hello!
hello!
```

Sample Solution

```
n = int(input("Enter an int: "))
word = input("Enter a word: ")
print()
for i in range(n):
    print(f'{i+1}. {word}')
```

Question 3 (q3_sol.py)

Write a program that asks the user to enter an int (called n) that is 0 or greater. Then, starting at 1, print the ints from 1 to n on the screen in a box as shown. Use the `print_in_box(word)` function from the previous question.

```
Enter an int: 3
```

```
+--+  
|1|  
+--+
```

```
+--+  
|2|  
+--+
```

```
+--+  
|3|  
+--+
```

Sample Solution

```
def print_inBars(word):  
    print(f'|{word}|')
```

```
def print_in_box(word):  
    bar = '+' + '-' * len(word) + '+'  
    print(bar)  
    print_inBars(word)  
    print(bar)
```

```
n = int(input("Enter an int: "))  
print()  
for i in range(1, n+1):  
    print_in_box(str(i))  
    print()
```

Question 4 (q4_sol.py)

Write a program that asks the user to enter an int that is greater than 1, and then prints a countdown from that int to 1, finally printing "Blastoff!".

```
Enter an int: 5
5
4
3
2
1
Blastoff!
```

Sample Solution

```
start = int(input("Enter an int: "))
for i in range(start):
    print(start - i)
print("Blastoff!")
```

Question 5 (q5_sol.py)

Write a program that asks the user to enter their age, and then tells them if they are old enough to vote in the BC provincial election. You must be 18 or older to vote in BC.

Sample run 1

```
Enter your age: 18
You are old enough to vote in the BC provincial election.
```

Sample run 2

```
Enter your age: 15
You are not old enough to vote in the BC provincial election.
```

Sample Solution

```
age = int(input("Enter your age: "))
if age >= 18:
    print("You are old enough to vote in the BC provincial election.")
else:
    print("You are not old enough to vote in the BC provincial election.")
```

Question 6 (q6_sol.py)

During the summer, a neighborhood has these rules for lawn watering:

- houses with even street numbers may water their lawn only on even-numbered days
- houses with odd street numbers may water their lawn only on odd-numbered days

Write a program that asks the user to enter a street number (like 503 or 1944) and a day number (Monday is 1, Tuesday is 2, ..., Sunday is 7), and then prints whether or not they house may water their lawn that day.

As an extra challenge, write and use a function called `can_water(street_number, day)` that returns `True` if the house may water their lawn that day, and `False` otherwise.

Sample run 1

```
Enter a street number: 503
Enter a day number: 1
You can water your lawn today.
```

Sample run 2

```
Enter a street number: 1944
Enter a day number: 2
You can water your lawn today.
```

Sample run 3

```
Enter a street number: 1944
Enter a day number: 3
You cannot water your lawn today.
```

Sample Solution

```
def can_water(street_number, day):
    if street_number % 2 == 0:
        return day % 2 == 0
    else:
        return day % 2 != 0

street_number = int(input("Enter a street number: "))
day = int(input("Enter a day number: "))
if can_water(street_number, day):
    print("You can water your lawn today.")
else:
    print("You cannot water your lawn today.")
```